

1. Scalar multiples and local bounds

Throughout these notes, functions are real-valued and defined on an open interval around a , except possibly at a . Conditions on x refer to their domains, or to the common domain when two functions are involved. We denote the domain of f by D . Distances may always be reduced to stay in these intervals.

Recall. The statement $\lim_{x \rightarrow a} f(x) = L$ means

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D : 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

We already know that limits are unique and that

$$f(x) \rightarrow L, \quad g(x) \rightarrow M \implies f(x) + g(x) \rightarrow L + M.$$

The notation here and below refers to $x \rightarrow a$.

Theorem. If $f(x) \rightarrow L$ and $c \in \mathbb{R}$, then $cf(x) \rightarrow cL$.

Proof. Let $\varepsilon > 0$. Since $\varepsilon/(|c| + 1) > 0$, the limit definition for f gives $\delta > 0$ such that $0 < |x - a| < \delta$ implies

$$|f(x) - L| < \frac{\varepsilon}{|c| + 1}.$$

For these x ,

$$|cf(x) - cL| = |c| |f(x) - L| \leq (|c| + 1) |f(x) - L| < \varepsilon.$$

The same choice also covers $c = 0$.

Difference law. Since $f - g = f + (-1)g$, the sum law and scalar-multiple law give $f(x) - g(x) \rightarrow L - M$.

Lemma (local boundedness). If $f(x) \rightarrow L \in \mathbb{R}$, then there are numbers $r > 0$ and $B > 0$ such that

$$0 < |x - a| < r \implies |f(x)| < B.$$

One can take $B = |L| + 1$.

Proof. Apply the definition with $\varepsilon = 1$ to find $r > 0$ such that $|f(x) - L| < 1$ for $0 < |x - a| < r$. Then the triangle inequality gives

$$|f(x)| \leq |f(x) - L| + |L| < 1 + |L| = B.$$

This only asserts a bound near a , with a excluded. It makes no assertion about the values far away or at a itself.

2. Products

Theorem. If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then

$$\lim_{x \rightarrow a} f(x)g(x) = LM.$$

The following identity expresses the product error in terms of the errors for the two factors:

$$f(x)g(x) - LM = f(x)(g(x) - M) + M(f(x) - L).$$

Thus

$$|f(x)g(x) - LM| \leq |f(x)| |g(x) - M| + |M| |f(x) - L|.$$

The first coefficient depends on x . Local boundedness gives a fixed number to use in its place.

Proof. Choose $r > 0$ such that $|f(x)| < B = |L| + 1$ whenever $0 < |x - a| < r$. Let $\varepsilon > 0$. The two assumed limits give $\delta_g, \delta_f > 0$ with

$$\begin{aligned} 0 < |x - a| < \delta_g &\implies |g(x) - M| < \frac{\varepsilon}{2B}, \\ 0 < |x - a| < \delta_f &\implies |f(x) - L| < \frac{\varepsilon}{2(|M| + 1)}. \end{aligned}$$

Set $\delta = \min\{r, \delta_g, \delta_f\} > 0$. For every x with $0 < |x - a| < \delta$, all three estimates hold, and

$$\begin{aligned} |f(x)g(x) - LM| &\leq |f(x)| |g(x) - M| + |M| |f(x) - L| \\ &\leq B|g(x) - M| + (|M| + 1)|f(x) - L| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

This proves the product law. The minimum makes the bound on $f(x)$ and both error estimates hold at the same x . No assumption that L or M is nonzero was needed.

Powers and polynomials. Repeated use of the product law gives

$$f(x)^n \rightarrow L^n \quad (n = 1, 2, 3, \dots).$$

Indeed, the case $n = 1$ is given, and multiplying the result for n by f proves the result for $n + 1$. Since $\lim_{x \rightarrow a} x = a$, constants, scalar multiples and finite sums now give

$$\lim_{x \rightarrow a} P(x) = P(a)$$

for every polynomial P .

3. Reciprocals and quotients

Lemma. If $g(x) \rightarrow M \neq 0$, then for some $r > 0$,

$$0 < |x - a| < r \implies |g(x)| > \frac{|M|}{2} > 0.$$

Proof. Apply the limit definition with $\varepsilon = |M|/2$ to make $|g(x) - M| < |M|/2$. Since $|M| \leq |M - g(x)| + |g(x)|$, we obtain

$$|g(x)| \geq |M| - |g(x) - M| > \frac{|M|}{2}.$$

In particular, $g(x) \neq 0$ throughout this punctured neighborhood.

Theorem. If $g(x) \rightarrow M \neq 0$, then $1/g(x) \rightarrow 1/M$.

Proof. In the neighborhood from the lemma,

$$\left| \frac{1}{g(x)} - \frac{1}{M} \right| = \frac{|g(x) - M|}{|g(x)||M|} \leq \frac{2}{|M|^2} |g(x) - M|.$$

Given $\varepsilon > 0$, choose $\delta_1 > 0$ such that

$$0 < |x - a| < \delta_1 \implies |g(x) - M| < \frac{\varepsilon |M|^2}{2}.$$

For $\delta = \min\{r, \delta_1\}$, the reciprocal is defined whenever $0 < |x - a| < \delta$, and the displayed estimate gives error less than ε . This proves the theorem.

Quotient law. If also $f(x) \rightarrow L$, the product law gives

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0.$$

In particular, for polynomials P, Q with $Q(a) \neq 0$, $\lim_{x \rightarrow a} P(x)/Q(x) = P(a)/Q(a)$.

Example

$$\lim_{x \rightarrow 2} \frac{x^3 - 1}{x^2 + 1} = \frac{8 - 1}{4 + 1} = \frac{7}{5}.$$

The polynomial limits exist and the denominator limit is $5 \neq 0$.

A zero denominator limit. The quotient law gives no conclusion when $M = 0$. For $x \neq 0$, $x/x = 1$ and $x^2/x = x$, so their limits at zero are 1 and 0. In both expressions the numerator and denominator tend to zero; those two facts do not determine the quotient limit. A quotient with no finite limit will appear next.

4. Non-existing limits

For a fixed candidate L , the assertion $\lim_{x \rightarrow a} f(x) = L$ is false exactly when

$$\exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in D : 0 < |x - a| < \delta \quad \text{and} \quad |f(x) - L| \geq \varepsilon_0.$$

To prove that no finite limit exists, we must establish this for *every* candidate $L \in \mathbb{R}$:

$$\boxed{\forall L \in \mathbb{R} \exists \varepsilon_0 > 0 \forall \delta > 0 \exists x \in D : 0 < |x - a| < \delta, \quad |f(x) - L| \geq \varepsilon_0.}$$

The number ε_0 may depend on L , but must be fixed before δ . The point may depend on both L and δ .

Example

The function $f(x) = |x|/x$, defined for $x \neq 0$, has no limit as $x \rightarrow 0$. Its values are

$$f(x) = \begin{cases} -1, & x < 0, \\ 1, & x > 0. \end{cases}$$

Proof. Fix any candidate $L \in \mathbb{R}$ and take $\varepsilon_0 = 1$. Given any $\delta > 0$, choose a point as follows.

- If $L \geq 0$, take $x = -\delta/2$. Then $0 < |x| < \delta$, while

$$|f(x) - L| = |-1 - L| = 1 + L \geq 1.$$

- If $L < 0$, take $x = \delta/2$. Then $0 < |x| < \delta$, while

$$|f(x) - L| = |1 - L| = 1 - L > 1.$$

For every candidate we have fixed $\varepsilon_0 = 1$ and found a point violating the required error bound in every punctured neighborhood. Therefore no finite limit exists.

The two values 1 and -1 are distance 2 apart. For any L ,

$$2 = |1 - (-1)| \leq |1 - L| + |-1 - L|,$$

so at least one of the two errors is at least 1. Both values occur arbitrarily close to zero. This observation also applies to the next example.

Assigning a value at zero cannot change the conclusion, because the definition always excludes $x = 0$.

5. Oscillation

Example

The function $f(x) = \sin(1/x)$, defined for $x \neq 0$, has no finite limit as $x \rightarrow 0$. We will find values 1 and -1 arbitrarily close to zero, even with $x > 0$.

The trigonometric identities we need are

$$\sin\left(\frac{\pi}{2} + 2\pi n\right) = 1, \quad \sin\left(\frac{3\pi}{2} + 2\pi n\right) = -1 \quad (n = 1, 2, 3, \dots).$$

To get these as function values, solve for x in the angle $1/x$.

Proof. Let $L \in \mathbb{R}$ be any candidate and set $\varepsilon_0 = 1$. Given any $\delta > 0$, choose a positive integer n so large that

$$2\pi n > \frac{1}{\delta}.$$

Such an integer exists because the positive integers have no upper bound. Define the two points

$$u = \frac{1}{\pi/2 + 2\pi n}, \quad v = \frac{1}{3\pi/2 + 2\pi n}.$$

Both denominators exceed $1/\delta$, so

$$0 < u < \delta, \quad 0 < v < \delta, \quad f(u) = 1, \quad f(v) = -1.$$

If $L \leq 0$, use $x = u$; then $|f(x) - L| = 1 - L \geq 1$. If $L > 0$, use $x = v$; then $|f(x) - L| = 1 + L > 1$.

Thus every candidate L fails the definition with $\varepsilon_0 = 1$, for every $\delta > 0$. There is no finite limit at zero.

Here $|f(x)| \leq 1$ everywhere in its domain, but the limit still fails to exist. Local boundedness is necessary for a finite limit, as proved in Section 1; it is not sufficient.

Reading. Stewart, *Calculus: Early Transcendentals*, 9th ed., §2.3 for limit laws and §2.4 for the precise definition.